

RISHI DADLANI

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EDUCATION

University of Pennsylvania, *College of Arts and Sciences*

Aug. 2023 – May 2027 (Exp.)

M.A. Mathematics, B.A. Mathematics

Philadelphia, PA

Minors: Computer Science, Statistics

Math GPA: 3.995/4.0 — Cumulative Undergraduate GPA: 3.92/4.0

RESEARCH INTERESTS

Differential & Riemannian Geometry, Geometric Analysis, PDE, Probability Theory.

SELECTED COURSEWORK

Analysis

PDE Theory I & II (PhD)

Functional Analysis (PhD)

Measure Theory (PhD)

Complex Analysis (PhD)

Probability Theory (PhD)

Topology/Geometry

Riemannian Geometry (PhD)

Smooth Manifolds (PhD)

Algebraic Topology (PhD)

Curves and Surfaces (Master's)

Point-Set Topology (Master's)

Algebra and Misc.

Groups, Rings, Modules (Master's)

Fields & Galois Theory (Master's)

Advanced Linear Algebra

Combinatorics and Graph Theory

Dynamical Systems (Seminar)

RESEARCH AND PAPERS

A Linking Integral Result

University of Pennsylvania

Advisor: Prof. Dennis DeTurck

– TBD

Aug. 2026 – May 2027 (Exp.)

Philadelphia, PA

Exponential DLPP Conditioned on Upper Deviation Event

University of Michigan REU

Advisor: Prof. Jinho Baik

– In Progress

June 2026 – Aug. 2026

Ann Arbor, MI

Collective Action as a Dynamical System

NSF Center for Analysis and Prediction of Pandemic Expansion (APPEX)

Advisor: Prof. Nina Fefferman

– Co-first author on *Classification of dynamics for model of planned behavior*: arxiv.org/abs/2601.11804.

– Analyzed a hybrid system modeling collective action; proved partial classification for n -individual case and for $n = 2$ proved a full classification of parameter space, verified this numerically.

May 2025 – Present

Remote

PROJECTS

Numerical Analysis of a Boltzmann Transport Equation

Texas A&M University PDE Summer Program

Advisor: Prof. Jean-Luc Guermond

– Proved energy estimates for a radiation transport PDE and implemented Streamline Upwind Petrov-Galerkin (SUPG), Richardson iteration, and a Poisson operator preconditioner informed by asymptotic expansion. Code available [here](#). Slides [here](#).

– Attended lectures on time-stepping methods, finite differences, finite elements.

May 2026

College Station, TX

Quillen's Q -Construction

University of Pennsylvania

Advisor: Prof. Ted Chinburg

– Defined geometric realization, classifying space, established equivalence between Grothendieck group completion of $\pi_0(M)$ and $\pi_0(\Omega BM)$.

– Reconstructed Quillen's definition of higher K -groups, $K_n^Q(R) := \pi_n(KR)$ via group completion of topological monoid $M_R = \coprod_P B\text{Aut}(P)$. Verified agreement with K_0, K_1, K_2 for rings.

March – June 2025

Philadelphia, PA

SELECTED READING PROJECTS

Characteristic Classes | Mentored by Prof. Dennis DeTurck

Summer 2026

– Stiefel-Whitney classes via Milnor & Stasheff; Thom, Euler, Chern classes and global angular form via Bott & Tu. Chern-Weil theory, Mathai-Quillen form, and Gauss-Bonnet-Chern Theorem via Weiping Zhang's book.

- Riemannian Geometry and Complex Manifolds** | Mentored by Prof. Dennis DeTurck Spring 2026
 – Metrics, connections, geodesics, normal coordinates, Riemann/Ricci curvature; almost complex structures, complex submanifolds, holomorphic bundles via Lee’s texts.
- Foliations & Integrability of Distributions** | Mentored by Profs. D. DeTurck, H. Gluck Winter 2025
 – Reeb foliation of S^3 , Denjoy’s results on circle diffeomorphisms, Frobenius theorem via Camacho and Lins Neto.
- Lie Groups and Lie Algebras** | Mentored by Prof. Dennis DeTurck Fall 2025
 – Coverings, adjoint representation, nilpotent/semisimple Lie algebras, structure theorems, and Cartan’s criteria.
- Applied Topology** | Mentored Dr. Shreya Arya Fall 2025
 – Read papers by R. Ghrist on Euler and Hadwiger integration and misc. papers on manifold learning
- Riemann Surfaces** | Mentored by Prof. Dennis DeTurck Summer 2025
 – Monodromy, Riemann–Hurwitz, classification of complex tori, and Riemann–Roch via Donaldson’s book.

TEACHING

- Head TA (EAS 5160 & 5170) & TA (ESE 3010)** Jan. 2026 – Aug. 2026
University of Pennsylvania
 – Held weekly problem sessions and office hours, graded weekly problem sets for ESE 3010.
 – Supervised and delegated workload throughout course staff for EAS 5160 & 5170.
- Course Developer: Linear Algebra (EAS 5170)** June 2025 – Jan. 2026
University of Pennsylvania
 – Worked with Professor Santosh Venkatesh to plan course content, design problem sets and exams, and ensure homogeneity for an engineering master’s linear algebra course.

TALKS

- Exponential DLPP Conditioned on Upper Deviation** | *YMC & University of Michigan REU* July 2026
 – Presented preliminary results on conditional exponential directed last passage percolation with flat initial condition at the Young Mathematicians Conference (Ohio State) and the University of Michigan REU. Slides [here](#).
- Radiation Transport (Boltzmann) Equation** | *Texas A&M PDE Summer Program* May 2026
 – 15-minute talk covering the May 2026 Boltzmann project described previously. Slides [here](#).
- Classification of Dynamics for a Model of Planned Behavior** | *Johns Hopkins MATRX* March 2026
 – Presented on my Summer 2025 research, motivated main results and outlined proofs. Slides [here](#).
- Functional Analysis Directed Reading Program** | *University of Pennsylvania* Dec. 2024
 – Introduced Banach and Hilbert spaces, basic results, Riesz representation theorem. Slides [here](#).
- Dynamical Systems Seminar** | *University of Pennsylvania* Sept. 2024 – Dec. 2024
 – Brin & Stuck’s *Dynamical Systems*; gave three one-hour lectures on mixing/ergodicity and circle dynamics.
- Lie Groups and Lie Algebras Class Presentation** | *University of Pennsylvania* April 2024
 – Introduced Lie groups and Lie algebras, Lie bracket of vector fields, Lie derivative, exponential map, basic representation theory and physical connections. Slides [here](#).

EXTRACURRICULARS & AWARDS

3× AIME | Penn Undergrad Math Society | Penn Symphony Orchestra | NAfME All-National Principal Flute

TECHNICAL SKILLS

Python (NumPy, Pandas, Matplotlib, Scikit-Learn, Seaborn, TensorFlow, Keras); Java; L^AT_EX